

COS30043

Interface Design and Development



Project

This is an individual project and is worth 40% of the total aggregated marks.

Project Overview

In this project, you are required to design and develop a full-stack web application using Vue.js for the frontend and an appropriate backend technology of your choice in order to implement fully functional and well-structured web application. The application must demonstrate a clear separation between the frontend and backend through the use of RESTful APIs.

The objective of this project is to evaluate your ability to design, implement, and integrate modern web application components, while exploring technologies and concepts beyond what are covered in lectures and tutorials.

Overall, this project will evaluate your ability to design, implement functional system and critically evaluate user interfaces that function effectively across multiple devices and platforms.

Functional Requirements

You must design and implement a web application consisting of at least 8 interconnected pages with any of the topics of your choice (e.g., e-commerce, booking system, event management, ordering system, etc.).

Example pages and not limited to:

- Landing / Main Page
- Product or Content Listing Page
- Product Detail Page
- Shopping Cart / Booking Page
- Registration Page/Login Page
- User Account / Dashboard Page
- Transaction or Activity History Page

Technical Requirements

1. Application of Interface Design Principles (ULO 1)

You must demonstrate the application of interface design concepts and standards such as:

- Contextual grouping of interface elements
- Consistent layout using a row–column grid system
- Clear visual hierarchy
- Consistent navigation design

2. Use of Contemporary Frameworks (ULO 2)

User interface must be developed using **Vue.js** version 3 and demonstrate appropriate use of modern frontend development techniques such as:

- Component-based architecture
- Reusable Vue components
- Use of Vue directives and conditional rendering
- Dynamic data binding
- Asynchronous data retrieval using APIs

3. Multi-Device Interface Design (ULO 3)

The application must support responsive design across multiple device types and implement mobile-first design principles to demonstrate the use of responsive layout techniques.

Screenshots demonstrating the application running on different device sizes (at least 3 sizes) must be included in the report. You must also describe how it is implemented.

4. Usability and Accessibility Evaluation (ULO 4)

You should evaluate your user interface using appropriate usability and accessibility techniques. This includes:

- Identifying potential usability issues
- Evaluating navigation efficiency
- Evaluating form usability
- Ensuring accessibility for different users

In your report, include a **brief usability evaluation** and propose improvements based on your findings.

5. Backend Integration

You should also develop the backend system to provide support to the front-end. This includes functional RESTful API with a database to handle CRUD operations.

To obtain Distinction or High Distinction marks, you must demonstrate independent exploration beyond what is taught in class. Examples include:

- Implementation of advanced Vue features

- Integration with external APIs
- Advanced component design patterns
- Improved accessibility features
- Performance optimisation techniques
- Advanced UI interactions or animations

You must clearly explain the additional techniques or tools you explored and how they enhanced the system in your report. Any additional/advanced techniques that are not included in the report will not be awarded marks.

Submission

You must submit the following components:

- Web application source codes (HTML, JavaScript, Backend Scripts)
- Project report. Refer to project report requirements for its content.
- Video presentation to be uploaded on YouTube

Project Report Requirements

Your project report should include the following sections:

- Cover page (Unit code, unit name, assessment, name, student ID, video presentation link, url if it's hosted remotely, project total number of pages)
- Introduction & system overview (2-3 pages)
- Wireframe design diagrams
- Screenshot of the web application in action
- Describe how you achieve a responsive UI (2-3 pages)
- Usability and accessibility evaluation and propose improvements (1-3 pages)
 - You should evaluate your user interface using appropriate usability and accessibility techniques. This includes:
 - Identifying potential usability issues
 - Evaluating navigation efficiency
 - Evaluating form usability
 - Ensuring accessibility for different
- Additional/advanced techniques implement which are not taught in class